SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA0618	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	GAYATHRI DEVI. M.S	**Reg. No.:**	192421279
Section / Batch:		**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
		GRAND TOTAL:	**100 Marks**

SECTION A – CONCEPT MAPPING

Code of Conduct
I GAYATHRI DEVI.M.S \ 192421279 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Gayathri Devi M.S

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

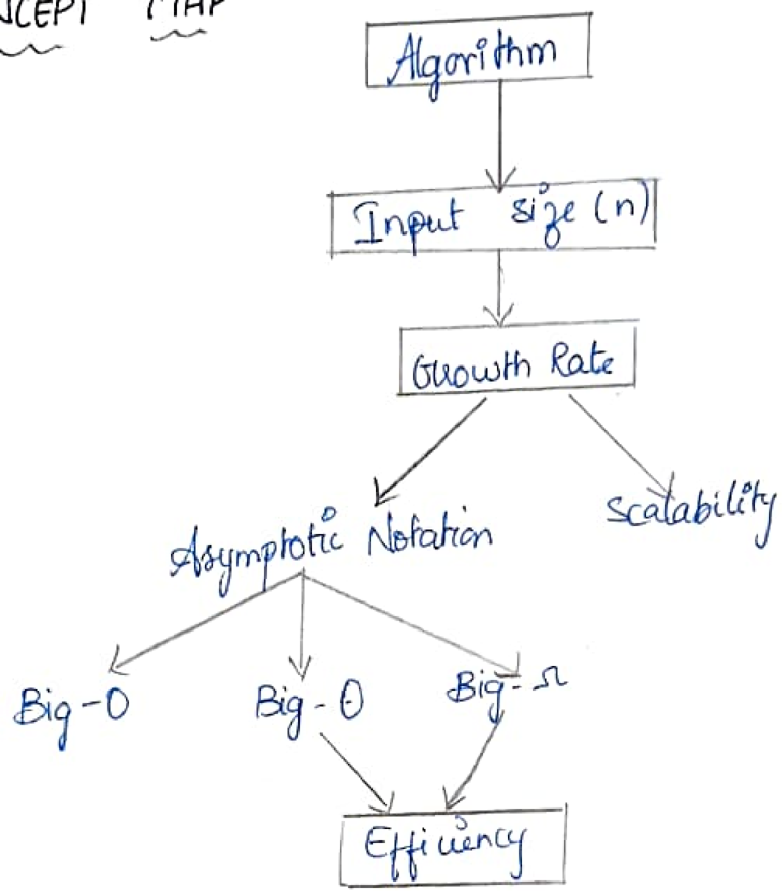
Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1	4	3	3	4	3	44
Q2	4	3	3	4	3	42
					Overall Total (100)	86

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

CO1 - AT-2 [CONCEPT MAPPING]

Algorithm Efficiency

CONCEPT MAP



EXPLANATION

⇒ Algorithm solves a problem

⇒ Input size (n) is the amount of data

⇒ As input size increase, growth rate increases

⇒ Growth rate decides the efficiency

⇒ Asymptotic notation measures performance

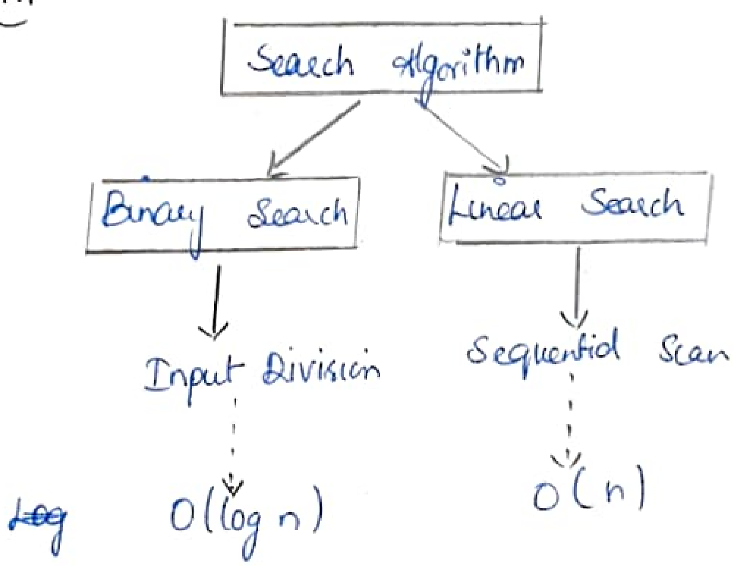
⇒ Scalability shows performance for large inputs

JUSTIFICATION:

As input size increases, efficiency depends on the growth rate. Better growth means better performance

2. Binary Search

CONCEPT MAP



EXPLANATION

- Binary Search divides the data into halves
- It has $O(\log n)$ complexity
- Linear Search checks one by one
- It has $O(n)$ complexity

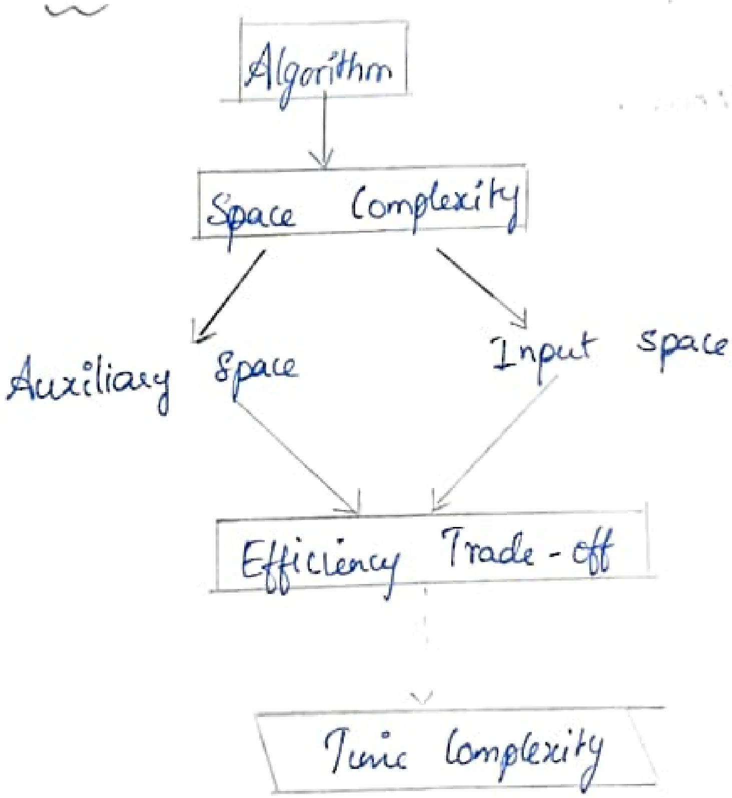
JUSTIFICATION:

Binary Search has a superior growth rate because it halves the search space at every step, while linear search examines every element. Therefore, Binary search performs much faster for larger dataset.

---

5) Space Complexity

CONCEPT MAP



EXPLANATION

- Space Complexity measures memory usage
- Auxiliary space is extra memory
- Input space stores input data

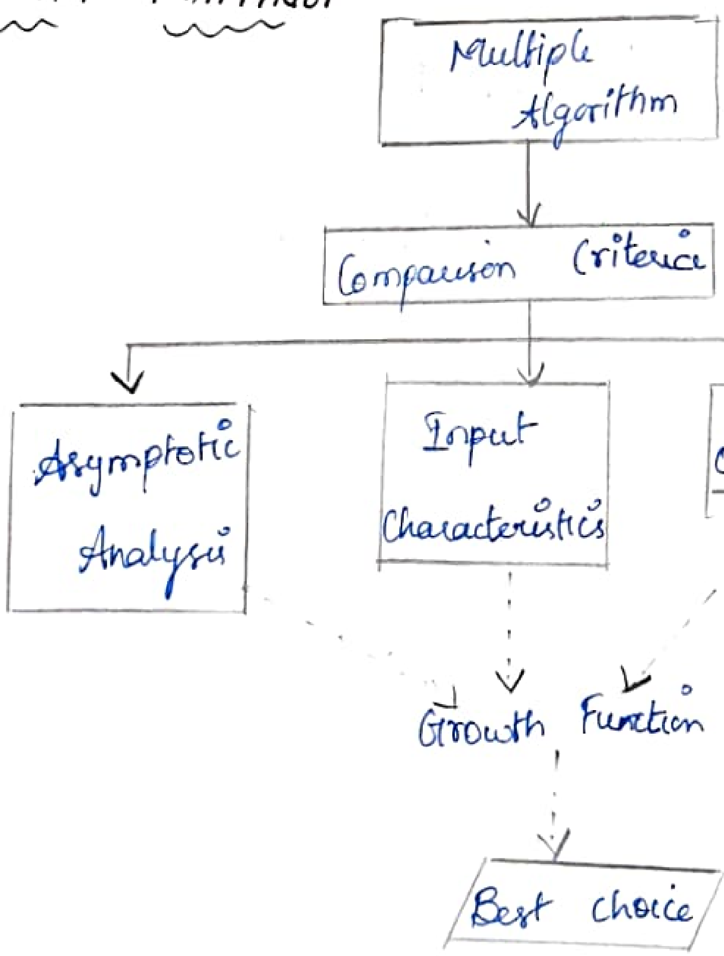
- Time and space affect each other

JUSTIFICATION:

Higher space usage is acceptable when it significantly reduce execution time. This time space trade-off helps design faster algorithm, especially in applications requiring high Performance

4. Algorithm Selection

CONCEPT MAPPING



EXPLANATION

- Many algorithm can solve one problem
- They are compared by growth, input and constraints
- The best one is selected

JUSTIFICATION

The concept map helps compare different algorithm systematically. By considering growth function, input characteristic and constraints, developers can choose the most suitable algorithm.

---

5. Complexity Classes

EXPLANATION

- Many algorithm can solve problem
- The input size affects algorithm performance.
- Complexity classes are:
  - → $O(1)$ - Constant
  - → $O(\log n)$ - Logarithmic
  - → $O(n)$ - Linear
  - → $O(n^2)$ - Quadratic
- Asymptotic Notation is used to represent these complexity

JUSTIFICATION:

Logarithmic and Linear algorithms are more scalable because their running time increases slowly as input grows. Quadratic algorithms become much slower for large inputs, making them less efficient. This concept map helps classify algorithm based on their order of growth & performance

CONCEPT MAPPING